



Chapter 5: NFT & Other Use Cases

By Maitri Hingu

Agenda

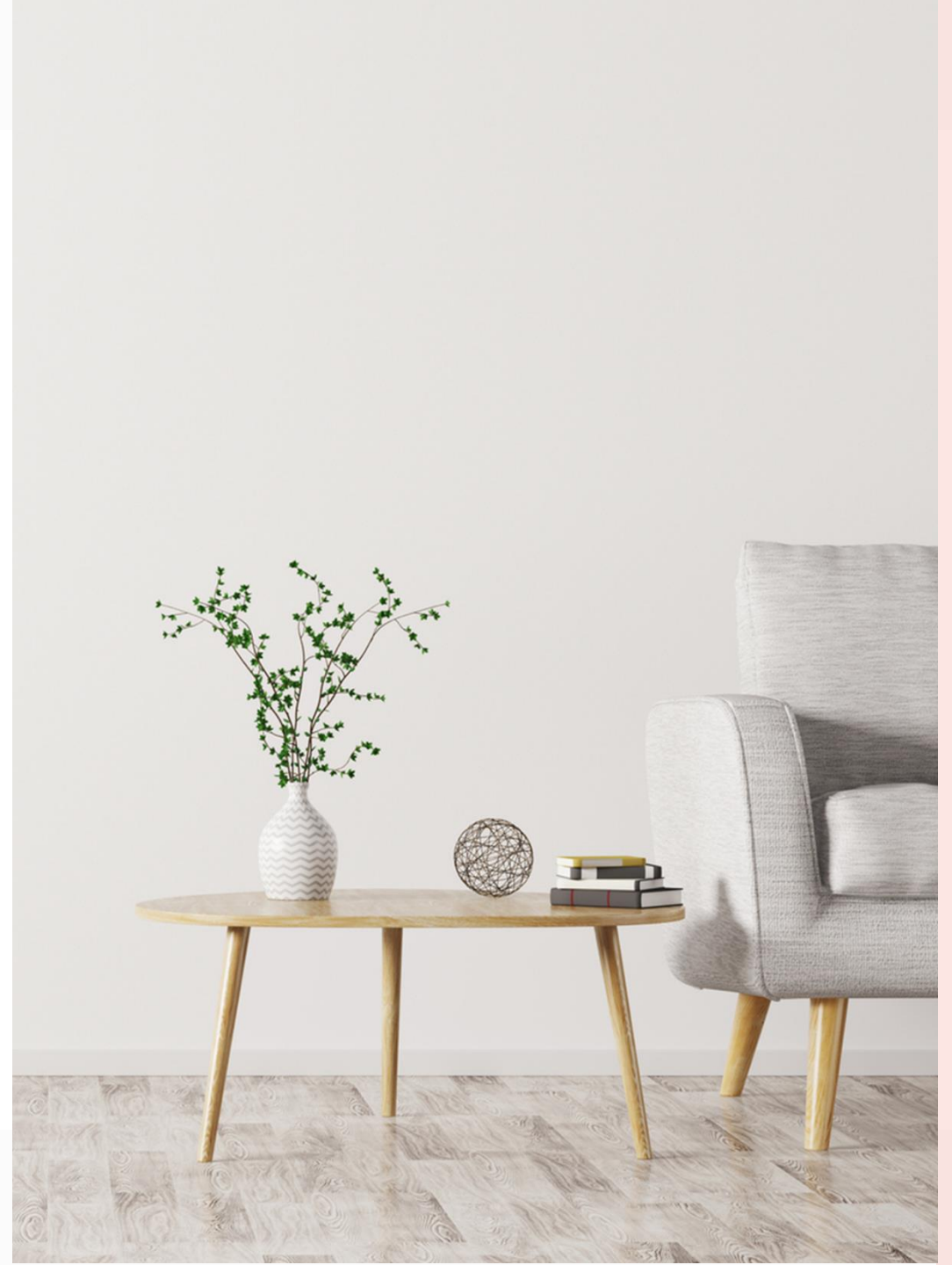


Introduction to NFT

- NFT v/s Cryptocurrency
- Types of NFT
- Creating, Buying & Selling NFT
- Impact of NFT on Environment
- NFT Usage and Rights

Use Cases

Introduction to NFT



Introduction to NFT

NFT stands for "Non-Fungible Token." To understand NFTs, let's break down the components:

- **Non-Fungible:**

"Fungible" means something that can be exchanged on a one-to-one basis. For instance, money is fungible. If you lend someone a \$10 bill and they return a different \$10 bill, you're still satisfied because the value is the same. Non-fungible, on the other hand, means something unique and not interchangeable. Each non-fungible item has unique properties that make it distinct from any other item.

- **Token:**

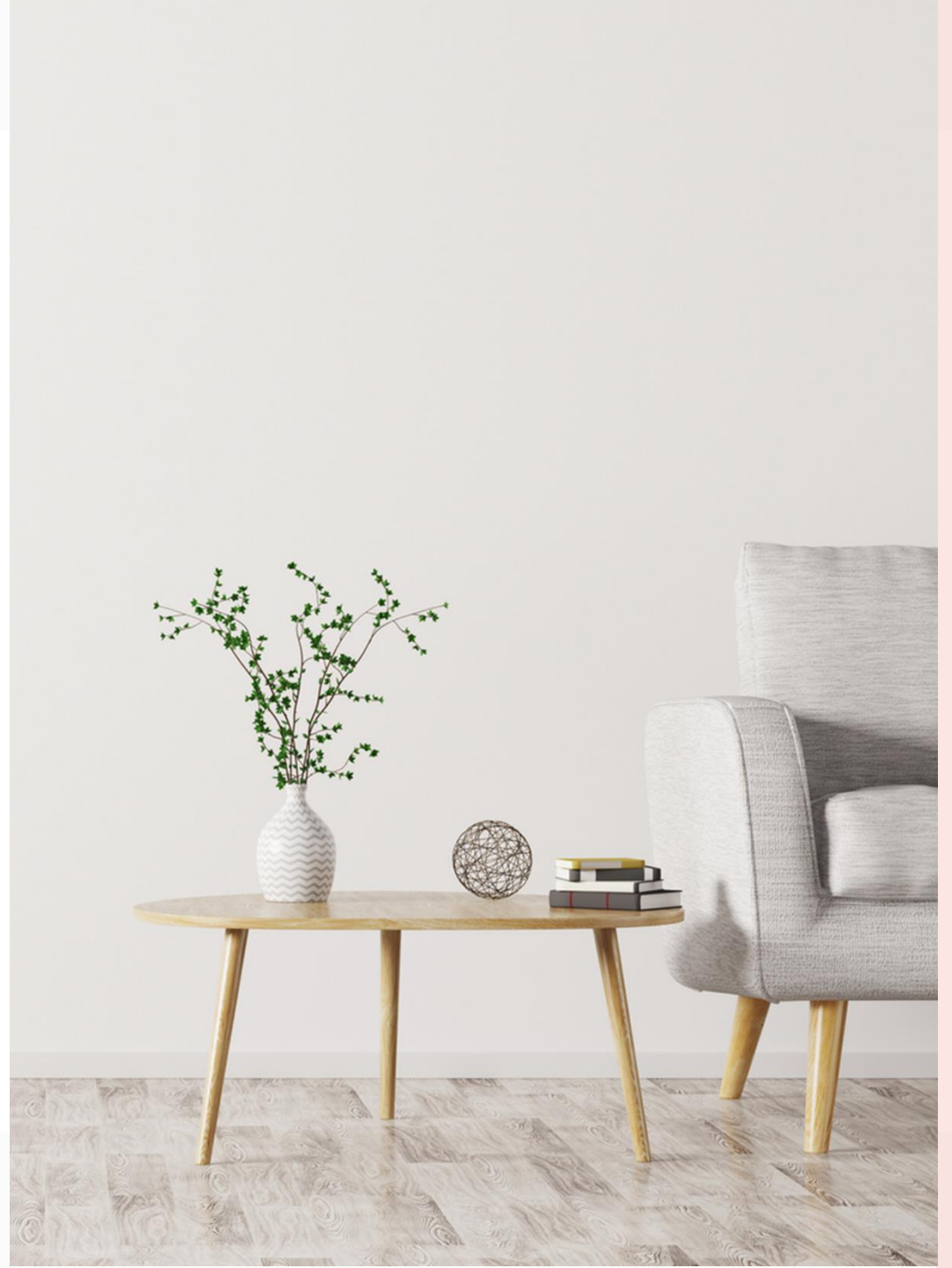
In the context of blockchain and cryptocurrencies, a token is a digital asset. It represents ownership of something and can be bought, sold, or traded like any other asset. Tokens can represent various things, including cryptocurrencies, digital art, collectibles, real estate, and more.

Introduction to NFT

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Now, combine these two concepts, and you have Non-Fungible Tokens (NFTs). NFTs are unique digital assets stored on a blockchain (usually Ethereum, but there are other blockchains supporting NFTs too). Each NFT has a distinct value and specific information that sets it apart from other tokens. This uniqueness is what makes NFTs valuable and in demand.

How NFT Works?



How NFT Works?

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NFTs operate on blockchain technology, which ensures their scarcity, authenticity, and ownership. Here's a basic overview of how they work:

- **Creation:**

An artist, creator, or entity mints an NFT using a smart contract on a blockchain platform like Ethereum. This smart contract contains the unique information about the NFT, such as its metadata, ownership details, and other relevant data.

- **Ownership:**

Once created, the NFT can be bought, sold, or traded like any other asset. Ownership of an NFT is recorded on the blockchain, providing transparency and immutability. Whoever owns the NFT has control over it and can transfer it to others as desired.

How NFT Works?

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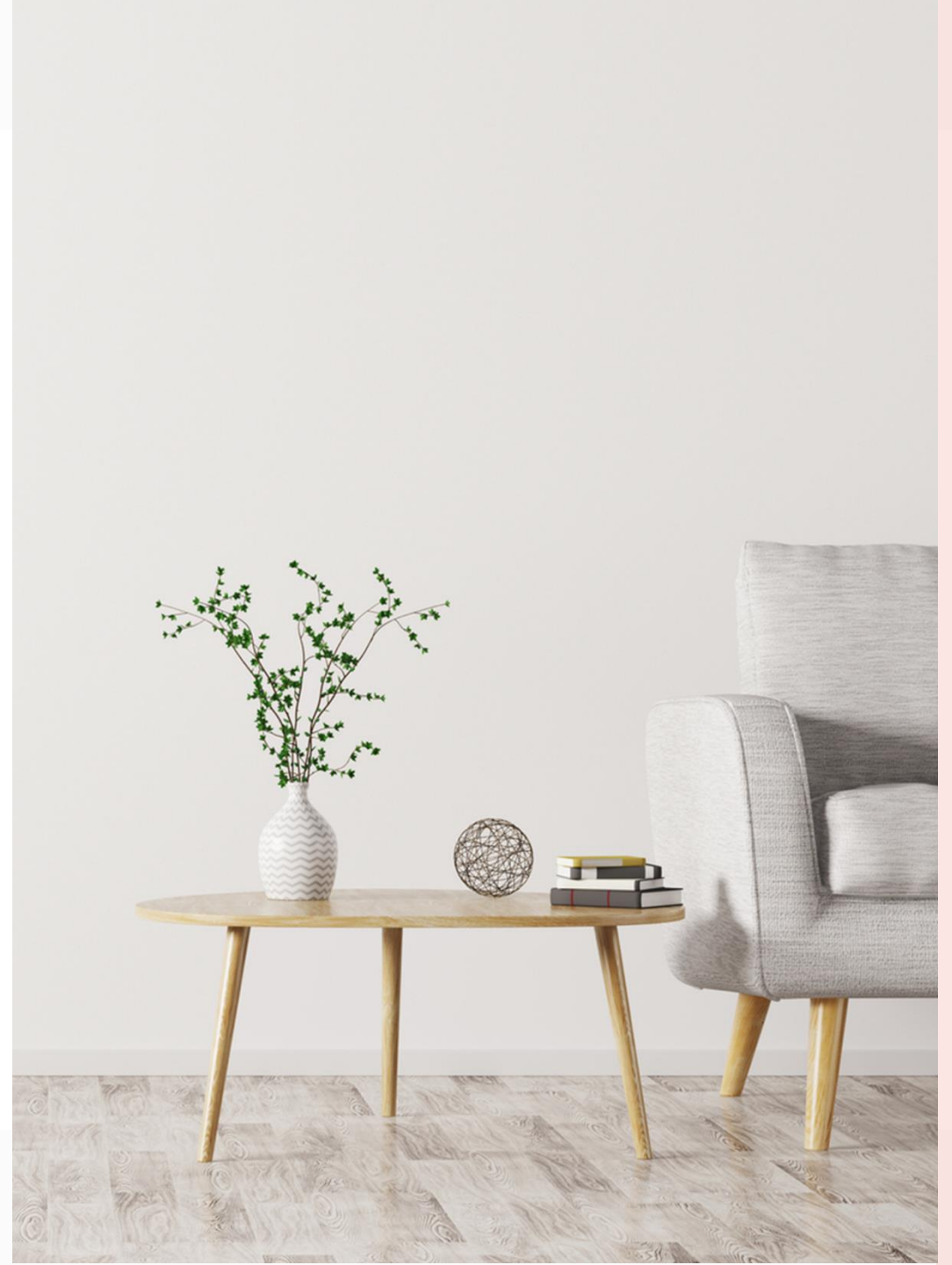
- **Metadata:**

NFTs often come with metadata that provides additional information about the asset. This metadata can include details about the creator, the creation date, the type of asset (artwork, music, video, etc.), and any other relevant information.

- **Provenance and Authenticity:**

The blockchain maintains a complete record of the NFT's ownership history, known as its provenance. This provenance ensures the authenticity and legitimacy of the NFT, as it verifies its origin and ownership over time.

Minting V/S Mining



Minting V/S Mining

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Minting	Mining
Process of creating and tokenizing unique digital assets on a blockchain.	Process by which new units of a cryptocurrency are created and added to the blockchain.
Involves the creation and recording of digital tokens, typically through the use of smart contracts and blockchain technology.	Involves validating and recording transactions on the blockchain network using computational power.
Primarily used for creating NFTs (Non-Fungible Tokens) or other unique digital assets.	Primarily used for securing the network by validating transactions and adding new blocks to the blockchain.

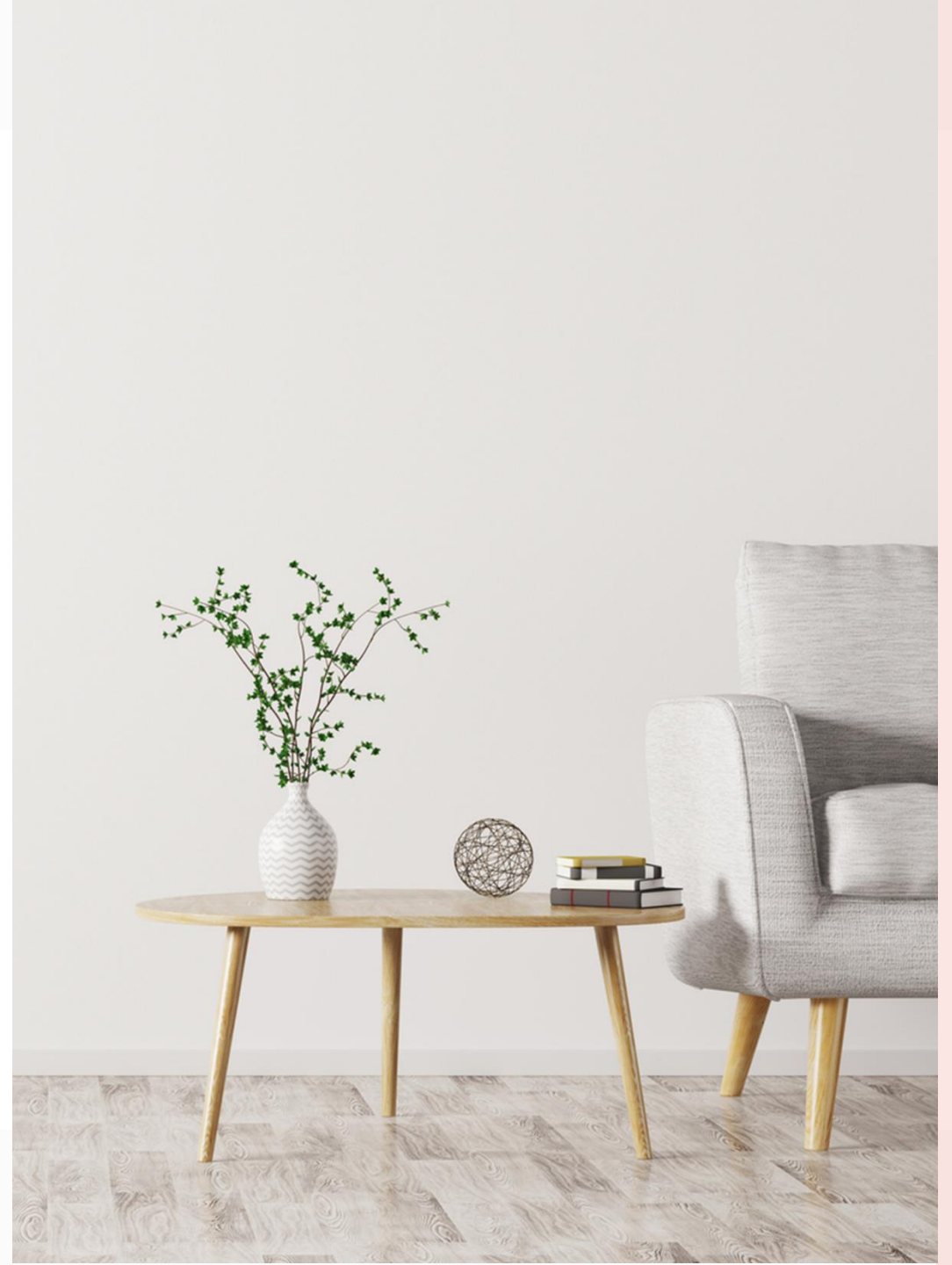
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Minting V/S Mining

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Minting	Mining
Tokens are created through the minting process and are typically associated with metadata such as ownership details, descriptions, and other relevant information.	Miners compete to solve complex mathematical puzzles in a proof-of-work (PoW) consensus mechanism.
Tokens are recorded on the blockchain to establish ownership, authenticity, and provenance of digital assets.	Miners who successfully validate a block of transactions are rewarded with newly minted cryptocurrency and transaction fees.

Types of NFT



Types of NFT

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Digital Art NFTs:

1. **Illustrations:** Digitally created drawings or illustrations, ranging from simple sketches to complex artworks.
2. **Paintings:** Digital paintings created using digital brushes and techniques, mimicking traditional painting styles.
3. **Animations:** Moving images or sequences created through digital animation techniques, including GIFs and short films.
4. **Digital Sculptures:** Three-dimensional digital models or sculptures created using software tools, often with intricate details and textures.

Types of NFT

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Collectibles:

1. **Virtual Trading Cards:** Digital cards representing characters, sports players, or fictional creatures, often with unique attributes and artwork.
2. **Characters:** Playable or non-playable characters within video games or virtual worlds, each with distinct abilities or traits.
3. **Toys and Figurines:** Digital representations of toys, action figures, or collectible figurines, often based on popular franchises or characters.
4. **Memorabilia:** Digital items associated with events, celebrities, or historical moments, such as concert tickets, autographs, or movie props.

Types of NFT

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Gaming Assets:

1. **In-Game Items:** Virtual items used within video games, including weapons, armor, potions, and cosmetic items like skins or costumes.
2. **Characters and Avatars:** Playable characters or avatars representing players within virtual worlds or multiplayer games.
3. **Virtual Currency:** Digital tokens or coins used as currency within video games for purchasing items, upgrades, or services.
4. **Land and Property:** Virtual land parcels or properties within virtual worlds or metaverses, often used for building structures or hosting events.

Types of NFT

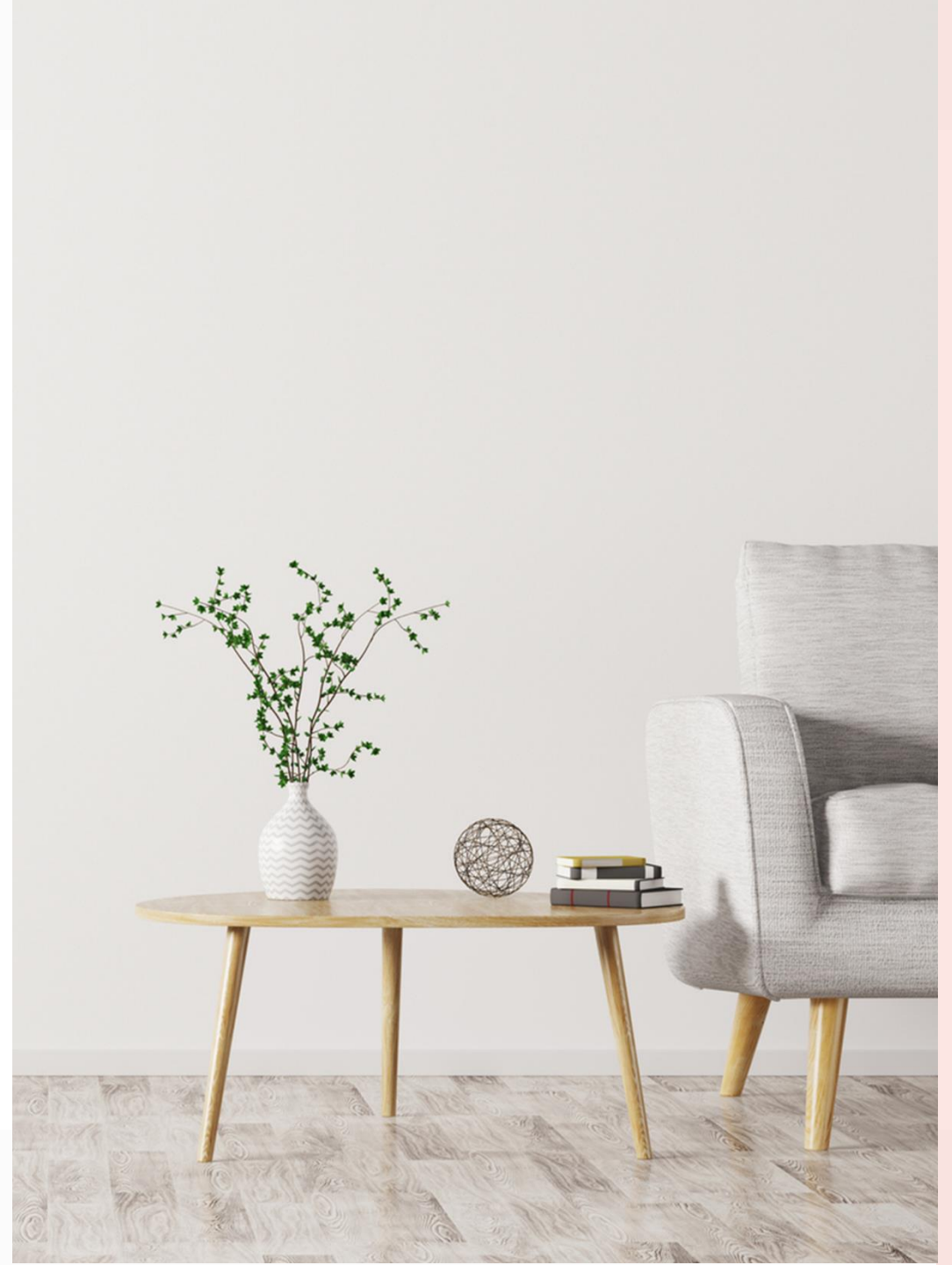
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Virtual Real Estate:

1. **Land Parcels:** Individual plots of virtual land within decentralized virtual worlds, often used for development or investment purposes.
2. **Virtual Buildings:** Digital structures or buildings constructed on virtual land, including houses, shops, and entertainment venues.
3. **Art Galleries and Museums:** Virtual spaces showcasing digital artworks, collectibles, and cultural artifacts within decentralized virtual environments.
4. **Event Spaces:** Virtual venues or arenas used for hosting events, conferences, concerts, and exhibitions within virtual worlds.

These sub-categories provide a more detailed breakdown of the various types of NFTs within the broader categories, highlighting the diverse range of digital assets and use cases within the NFT ecosystem.

Creating, Buying & Selling NFTs



Creating NFT

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Creating an NFT involves several steps, from setting up a cryptocurrency wallet to minting your digital artwork on a blockchain platform.

Below is a practical step-by-step guide:

Set Up a Cryptocurrency Wallet:

- Choose a cryptocurrency wallet that supports the blockchain platform you intend to use for minting your NFT. Popular options include MetaMask for Ethereum-based NFTs or Trust Wallet for Binance Smart Chain-based NFTs.
- Download and install the wallet app on your device.
- Follow the instructions to create a new wallet and securely store your private key or seed phrase. This key is crucial for accessing your wallet, so keep it safe and never share it with anyone.

Creating NFT

Acquire Cryptocurrency:

- Purchase cryptocurrency (such as Ethereum for Ethereum-based NFTs or BNB for Binance Smart Chain-based NFTs) through a cryptocurrency exchange like Coinbase, Binance, or Kraken.
- Transfer the cryptocurrency to your wallet address. Make sure you're sending it to the correct address and double-check the transaction details before confirming.

Choose a Marketplace or Platform:

- Decide where you want to mint and sell your NFT. Popular NFT marketplaces include OpenSea, Rarible, and Mintable for Ethereum-based NFTs, and platforms like BakerySwap for Binance Smart Chain-based NFTs.
- Create an account on the chosen platform and connect your cryptocurrency wallet to it. This allows you to interact with the platform and manage your NFTs.

Creating NFT

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Create Your Digital Art:

- Produce or prepare the digital artwork you want to mint as an NFT. Ensure that the artwork meets the platform's guidelines and specifications for file format, size, and other requirements.
- Save your artwork in a high-quality digital format.

Mint Your NFT:

- Navigate to the "Create" or "Mint" section of the chosen platform. Follow the platform's instructions to upload your digital artwork file. You may need to provide additional information such as a title, description, and tags for your NFT.
- Set the parameters for your NFT, including the number of copies (if applicable), pricing, and royalties.
- Confirm and finalize the minting process. This typically involves signing a transaction using your cryptocurrency wallet to pay for the minting fees.

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Creating NFT

Verify and Publish:

- Once the minting process is complete, your NFT will be published on the blockchain and available for sale on the marketplace.
- Verify that your NFT is displayed correctly and appears as intended. Make any necessary adjustments if needed.

Market and Sell Your NFT:

- Promote your NFT on social media, forums, and other channels to attract potential buyers.
- Monitor the marketplace for inquiries, bids, or offers on your NFT.
- Once a buyer purchases your NFT, the ownership transfer will be recorded on the blockchain, and you'll receive the corresponding cryptocurrency payment in your wallet.
- Remember to stay informed about any platform-specific guidelines, fees, and best practices to ensure a smooth experience throughout the NFT creation and selling process.

Buying NFT

Research and Choose a Marketplace:

- Research various NFT marketplaces such as OpenSea, Rarible, Foundation, and others.
- Consider factors like user interface, fees, community, and supported blockchain networks.

Set Up a Wallet:

- Choose a cryptocurrency wallet that supports the blockchain network used by the marketplace (e.g., MetaMask for Ethereum-based NFTs).
- Create or import your wallet and securely store your private key or seed phrase.

Fund Your Wallet:

- Purchase cryptocurrency (e.g., Ethereum) from a cryptocurrency exchange.
- Transfer the purchased cryptocurrency to your wallet address.

Buying NFT

Browse and Find NFTs:

- Explore the marketplace to find NFTs you're interested in purchasing.
- Use filters, categories, and search tools to narrow down your options.

Place a Bid or Buy Now:

- Once you find an NFT you want to purchase, review its details, including price, description, and creator information.
- Decide whether to place a bid or buy it now (if available).

Confirm Transaction:

- If placing a bid, enter your bid amount and review the transaction details.
- If buying now, review the purchase price and confirm the transaction.

Buying NFT

Sign Transaction:

- Sign the transaction using your wallet to authorize the purchase.
- Pay attention to gas fees (transaction fees) when buying NFTs on Ethereum-based platforms.

Ownership Transfer:

- After the transaction is confirmed, ownership of the NFT is transferred to your wallet address.
- The NFT will now be visible in your wallet and the marketplace.

Selling NFT

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Create an Account:

- Sign up for an account on the NFT marketplace of your choice.
- Connect your cryptocurrency wallet to the platform.

Prepare Your NFT:

- Ensure your digital artwork or asset meets the platform's guidelines and specifications.
- Prepare high-quality images or files for uploading.

Mint Your NFT:

- Navigate to the minting section of the marketplace and follow the instructions to mint your NFT.
- Upload your digital asset, add metadata (title, description, tags), and set pricing and royalties.

Selling NFT

List Your NFT for Sale:

- Once minted, your NFT will be available in your account inventory.
- Set a sale price or enable bidding, and choose whether to list it as an auction or fixed price.

Promote Your NFT:

- Share your NFT listing on social media, forums, and other platforms to attract potential buyers.
- Engage with the community and participate in relevant discussions to increase visibility.

Respond to Offers and Inquiries:

- Monitor your NFT listings for offers, bids, or inquiries from potential buyers.
- Respond promptly and professionally to inquiries and negotiate offers if necessary.

Selling NFT

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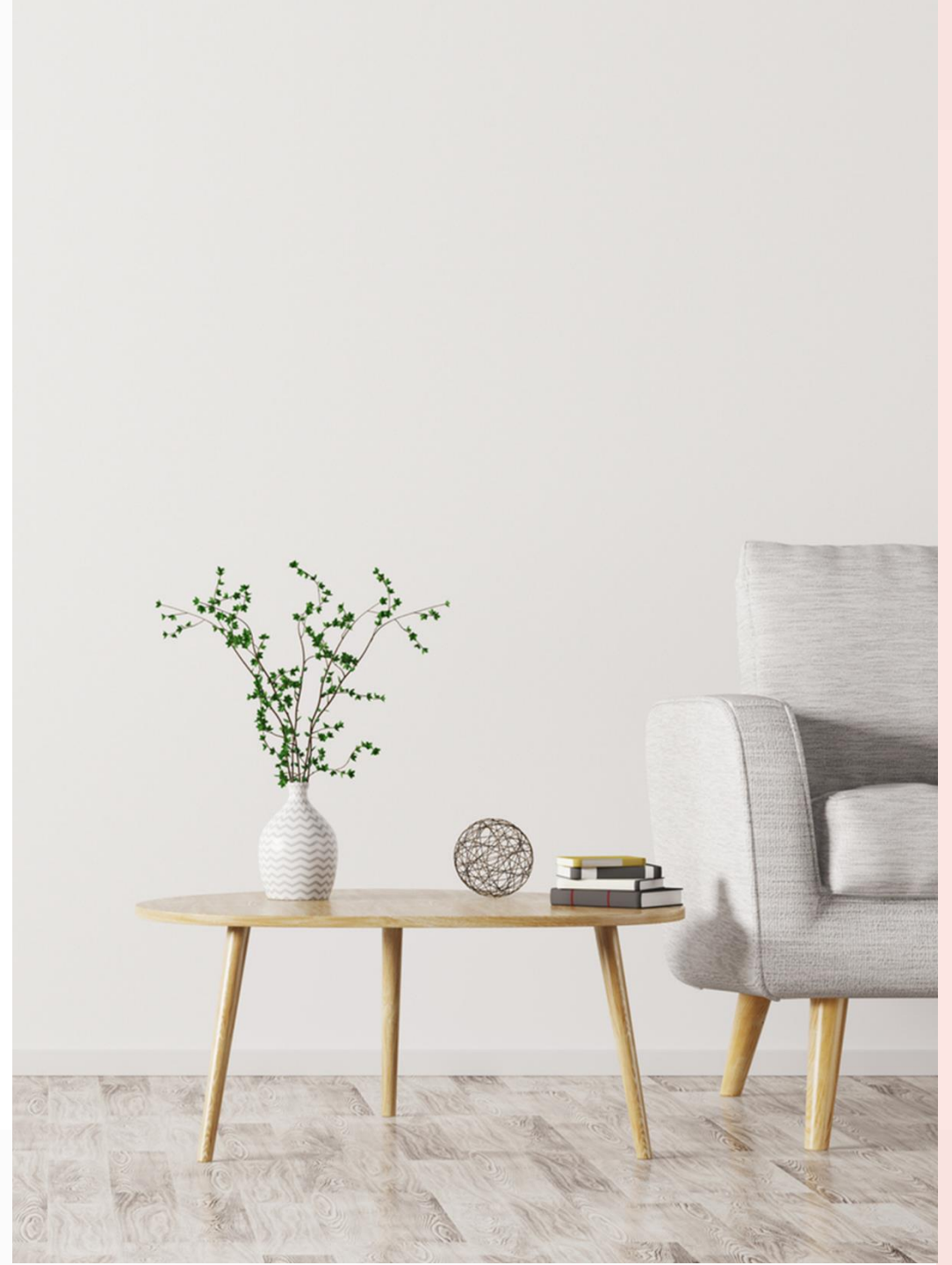
Complete the Sale:

- Once a buyer purchases your NFT, confirm the sale and transfer ownership to the buyer's wallet.
- Ensure the buyer has completed the payment before transferring ownership.

Withdraw Funds:

- Withdraw any earned cryptocurrency from the platform to your wallet or exchange for fiat currency if desired.

Criticisms, Challenges & Impact of NFT on Environment



Criticisms & Challenges

While NFTs offer numerous opportunities, they also face criticisms and challenges, including:

- **Environmental Concerns:**

The energy consumption associated with blockchain technology, particularly proof-of-work consensus mechanisms like Ethereum's, has raised environmental concerns due to its carbon footprint.

- **Market Speculation and Volatility:**

The NFT market has experienced significant speculation and volatility, leading to concerns about inflated prices, market bubbles, and potential financial risks for investors and creators.

- **Copyright and Ownership Issues:**

NFTs raise questions about copyright and ownership rights, especially regarding digital art and intellectual property. Determining the rightful owner of an NFT and resolving disputes can be challenging.

- **Scams and Fraud:**

The decentralized nature of blockchain technology can make it susceptible to scams, fraud, and security vulnerabilities. As a result, buyers and creators need to exercise caution and due diligence when participating in the NFT market.

Impact of NFT on Environment

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The impact of NFTs (Non-Fungible Tokens) on the environment has been a topic of discussion and concern, primarily due to the energy consumption associated with the blockchain technology used to create and trade NFTs.

Here are some key points to consider regarding the environmental impact of NFTs:

Blockchain Energy Consumption:

NFTs are often minted and traded on blockchain platforms like Ethereum, which primarily use proof-of-work (PoW) consensus mechanisms.

PoW requires significant computational power to validate transactions and secure the network, leading to high energy consumption.

Mining and validating transactions on PoW blockchains consume electricity, contributing to carbon emissions and environmental degradation.

Impact of NFT on Environment

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Carbon Footprint:

The energy consumption of PoW blockchains has led to concerns about their carbon footprint and contribution to climate change.

Studies have estimated the carbon footprint of cryptocurrencies and blockchain technology, with some indicating significant environmental impacts.

Criticism of NFTs:

NFTs have faced criticism for their environmental impact, especially as high-profile sales and auctions generate significant interest and transactions.

Critics argue that the environmental cost of creating and trading NFTs may outweigh their cultural or artistic value.

Impact of NFT on Environment

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Efforts to Address Environmental Concerns:

Some blockchain platforms are exploring alternative consensus mechanisms, such as proof-of-stake (PoS), which require less energy compared to PoW.

Initiatives like Ethereum 2.0 aim to transition Ethereum from PoW to PoS, potentially reducing its environmental footprint.

Artists, collectors, and platforms are increasingly considering the environmental impact of NFTs and seeking sustainable solutions.

Impact of NFT on Environment

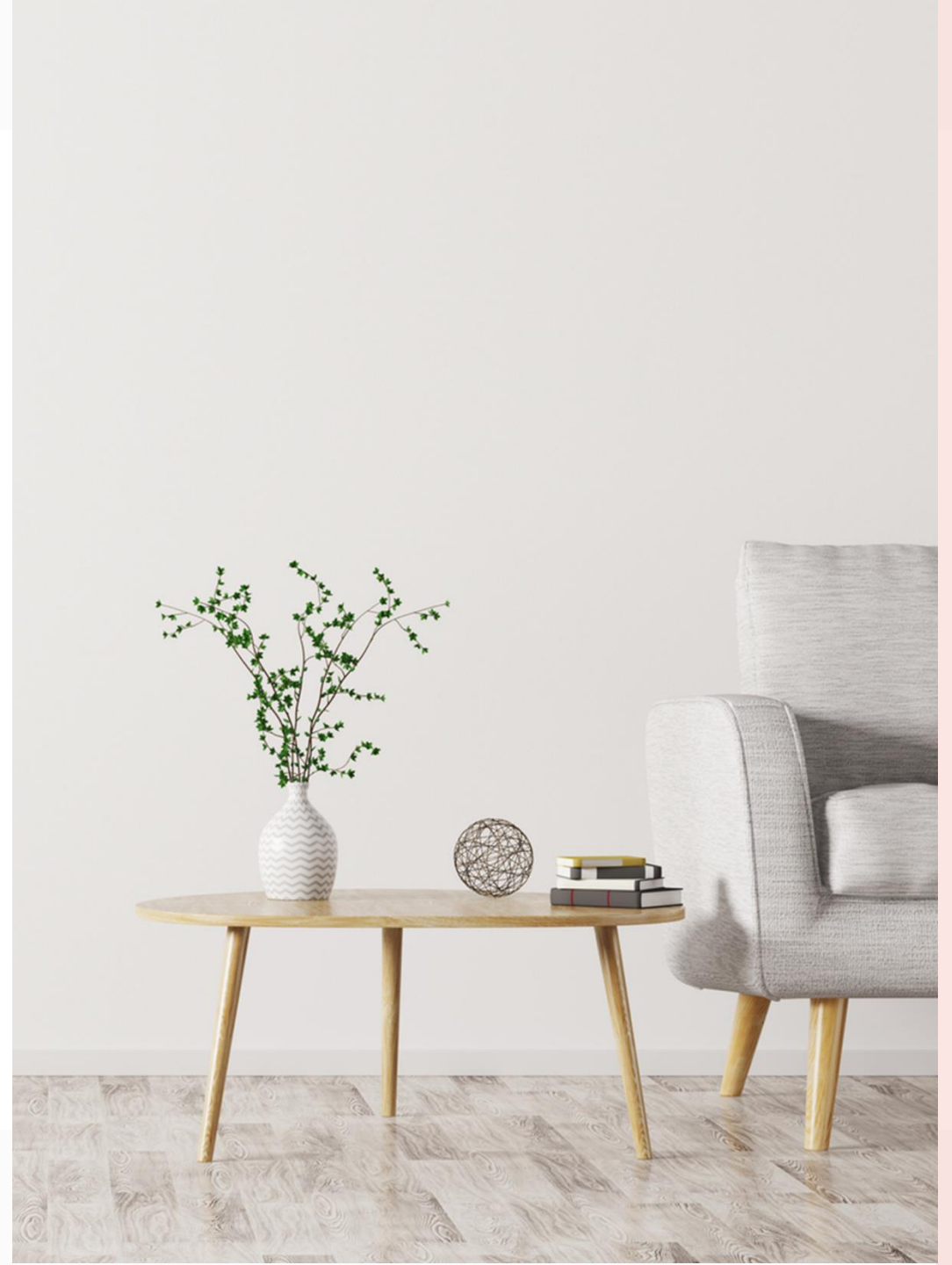
Carbon Offsetting and Sustainability Efforts:

Some NFT projects and platforms are implementing carbon offsetting measures or donating a portion of proceeds to environmental causes to mitigate their impact.

Additionally, there is growing interest in creating NFTs on blockchain networks with lower energy consumption, such as those utilizing PoS or other energy-efficient consensus mechanisms.

In summary, while NFTs have raised environmental concerns due to their association with energy-intensive blockchain technology, efforts are underway to address these concerns and promote sustainability within the NFT ecosystem. As the technology evolves and environmental awareness grows, stakeholders are exploring ways to minimize the environmental impact of NFTs while maximizing their cultural and economic value.

Use Cases of NFTs



NFTs have gained popularity across various industries due to their unique properties and potential applications. Some common use cases include:

- **Digital Art and Collectibles:**

NFTs enable artists to tokenize their digital artworks, creating a new market for digital art and collectibles. Artists can sell their work directly to collectors, who can then prove ownership through the blockchain.

- **Gaming and Virtual Worlds:**

NFTs are used in gaming and virtual worlds to represent in-game assets, such as skins, items, characters, and land. Players can buy, sell, and trade these assets both within and outside of the game environment.

- **Music and Entertainment:**

Musicians, filmmakers, and other content creators can tokenize their creations as NFTs, allowing fans to purchase and own unique digital memorabilia or access exclusive content.

- **Tokenization of Physical Assets:**

NFTs can represent ownership of physical assets, such as real estate, luxury goods, and intellectual property rights. This tokenization allows for fractional ownership, increased liquidity, and easier transfer of ownership.

Use Cases of NFTs

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IMPORTANT NOTE:

For use-cases of blockchain refer the google data and prepare it on your own. Real Estate, Insurance, Governance and other industries; this are the topics. Search on it and understand its offline working and then what good will be done if we use blockchain for it.

Thank You

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